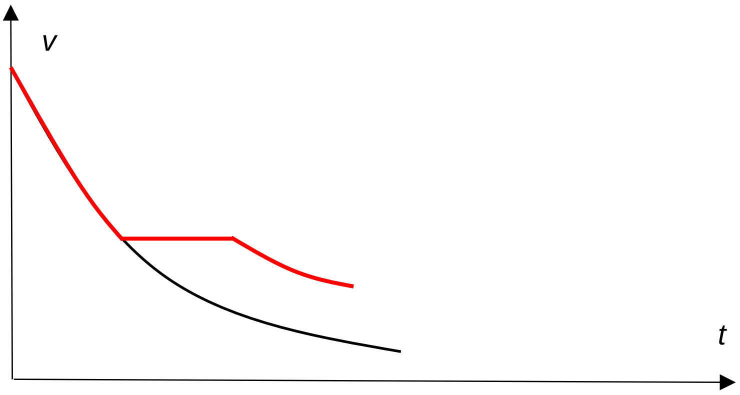
Q7		
(a)	The magnetic flux (linkage) is given by $\Phi = NBA = B(wx)$ where x is the distance AB has moved past P. Hence the induced emf is given using **Faraday's Law** by $E = \frac{d\Phi}{dt} = Bw\frac{dx}{dt}$ $Bw\frac{dx}{dt} = Bwv$	**B2**
(b)(i)	As AB moves from P towards Q, **magnetic flux linkage** over the area ABCD enclosed by the frame **increases** resulting in an **induced e.m.f.** generated in the frame **by Faraday's Law**. **By Lenz's Law**, an **induced current** will flow **such that it opposes the increase in magnetic flux linkage**. (current flows in anticlockwise direction) Consequently, a **magnetic force acts on AB towards the left**, causing it to slow. (Using Fleming's Left Hand Rule) <u>Alternative</u>: AB **cuts the magnetic flux** as it moves through PQ, resulting in an induced e.m.f. generated in the frame **by Faraday's Law**. There is **induced current** in the full circuit **in the anticlockwise direction**. (or electrons in the clockwise direction) Consequently, a **magnetic force acts on AB towards the left**, causing it to slow. B1 – why there is an induced emf B1 – how is the direction of induced current determined B1 – effect on AB/frame	**B1** **B1** **B1** **B1** **B1** **B1**
(b)(ii)	 B1 – same slope where the speed is the same as the original graph B1 – plateau B1 – shorter time to pass through the field, higher speed as it leaves the field	**B3**

Q8		